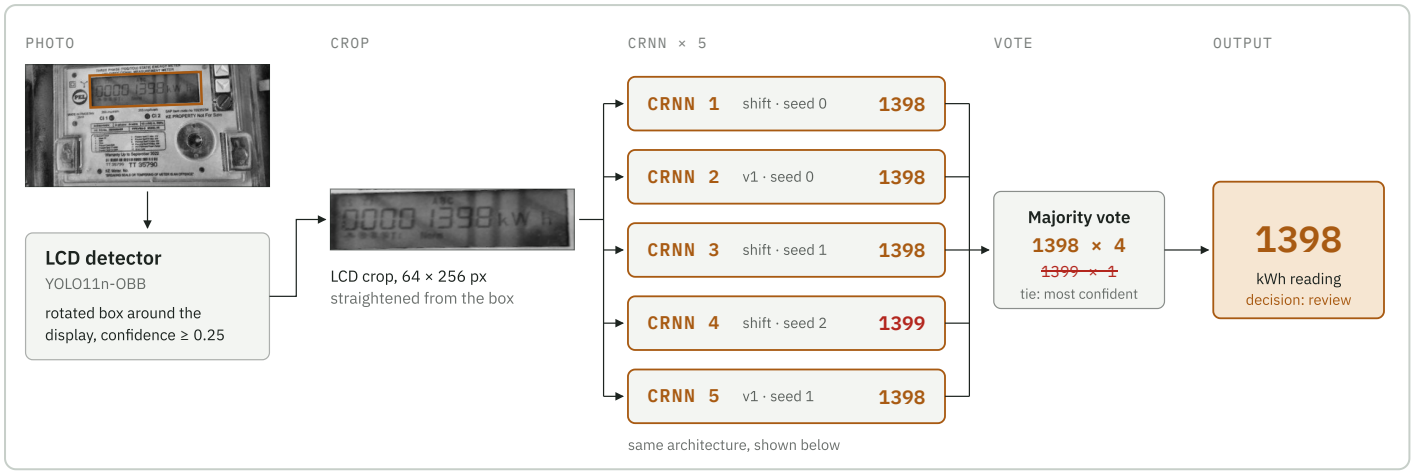
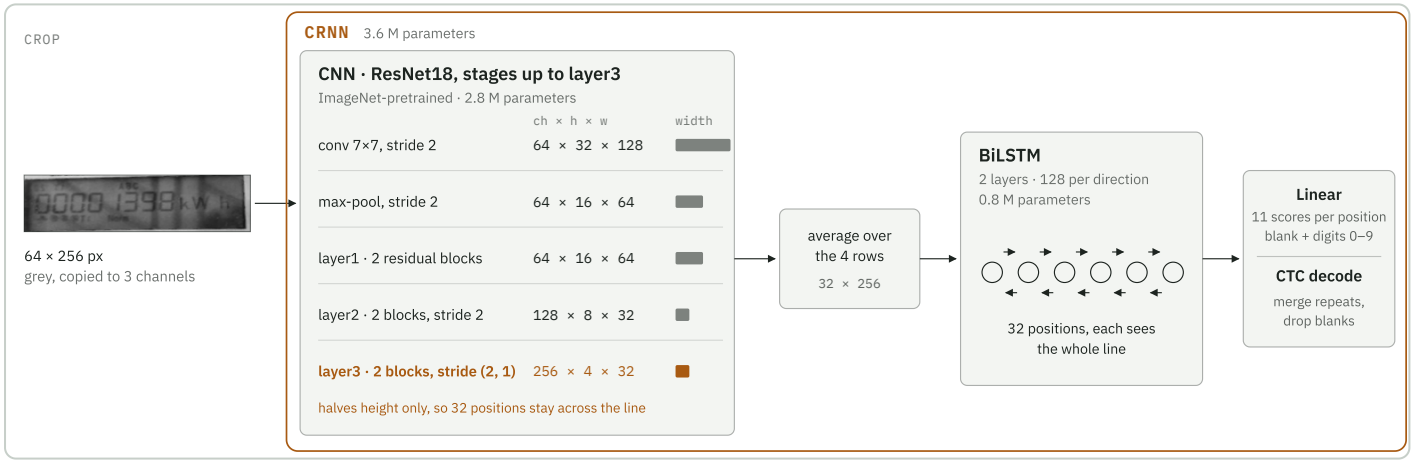
CV Pipeline



v1	training augmentation: random box shifts and rotation, brightness, contrast, gamma, blur, noise
shift	v1 plus moiré stripes, faded display, low resolution, JPEG artefacts and motion blur, to imitate poor photos
seed	random starting point of training; the same settings with a different seed give a slightly different model

Inside each CRNN



Reading a photo

WHAT THE READER DOES FOR EVERY PHOTO

1. **Blank check.** A photo with almost no edges (edge density below 0.03) stops here as *unreadable*.
2. **Detector.** The top rotated box at confidence ≥ 0.25; with no box the photo is *unreadable*.
3. **Photo quality.** Moiré, washed-out display, blur and glare are measured; the result feeds the decision.
4. **8 boxes.** The detected box plus 7 shifted copies: centre ±2% of the width and ±4% of the height, width × 0.97–1.08, height × 0.95–1.10, angle ±1.5°.
5. **4 versions** of each crop: original, contrast boost (CLAHE), moiré filter (frequency notch) and glare filled (bright spots painted in), each resized to 64 × 256. That makes 32 crops.
6. **Each CRNN** reads all 32. For each version, its confidence is the share of the 8 reads that match the unshifted read; it keeps the version with the highest confidence and that version's unshifted reading.
7. **Vote.** The reading given by most CRNNs wins; a tie goes to the most confident CRNN, then the lower number.



Decision

DECISION	RULE	NEXT STEP
`accept`	all 5 CRNNs give the same reading and their mean confidence is at least 7/8	use the reading
`review`	everything else	a person checks the reading, which is filled in
`unreadable`	blank photo, no display found, or a photo flagged for moiré or a blank display where at most 3 CRNNs agree	retake the photo

The example above: 4 of the 5 CRNNs read 1398, so the reading is 1398 and it goes to **review**.

Results

606 LABELLED TEST CROPS (KE-METER-READINGS TEST SPLIT) · FIVE CRNNS TRAINED ON 4,248 LABELLED CROPS, 50 EPOCHS, FINAL EPOCH

Five-model vote	**98.2%** exact (595 of 606, 95% CI 96.8–99.0%) · 99.6% of digits right
Single CRNNs	96.7–97.7% each
Full reader	98.0% exact (594 of 606) with the 4 versions, 8 reads and the decision
accept	580 crops (95.7%) · **577 correct, 99.5%**
review	25 crops (4.1%) · 17 correct
unreadable	1 crop (0.2%)
Wrong readings	12 in total: 9 go to review or unreadable, 3 are accepted
Validation	1,214 crops: five-model vote 96.8%
From full photos	the same 606 meters read from the original photos, with every step above (blank check, detector, our crop): **98.2%** (595 of 606); accept 95.9% of photos at 99.1% correct; display found on all 606
Shared meters	225 test crops come from meters also in training; they score 97.8%, the other 381 score 98.4%, so the overlap does not raise the result

Training

LCD detector · YOLO11n-OBB

Labels	350 rotated boxes drawn by hand, 1 class (`lcd`): 247 hard photos from the 6,284-photo corpus + 103 from the first labelled 1,000
Hard photos	ranked by low confidence of the previous detector (weight ×3), moiré, blur, strong tilt and unusual display size; missed or wrong boxes always included; one photo per meter
Split	281 train / 69 validation, grouped by meter serial
Start	`yolo11n-obb.pt`, pretrained
Length	100 epochs, batch 16, image size 512 (photos are 500 × 250); early-stopping patience 30, not triggered
Optimiser	Ultralytics `auto` → AdamW, lr 0.002, momentum 0.9 (runs under 10,000 steps; this one is 1,800); weight decay 0.0005
Schedule	3 warm-up epochs, then linear decay to 1% of the start rate
Loss weights	box 7.5, class 0.5, DFL 1.5, plus the rotated-box angle loss
Augmentation	mosaic (off for the last 10 epochs), rotation ±10°, translate ±10%, scale ±50%, HSV (hue 0.015, saturation 0.7, value 0.4); no flips, so digits are never mirrored
Run	mixed precision, seed 0, deterministic; about 8 minutes on an RTX 3050 laptop GPU
Model kept	best epoch, 92 (fitness 0.1 × mAP50 + 0.9 × mAP50-95 on the 69 validation boxes)
Result	validation precision 0.999, recall 1.000, mAP50 0.995, mAP50-95 0.946; on 100 separate hand-drawn test boxes, 93% correct at IoU ≥ 0.8 and none missed

CRNNs

DATA AND RUNS	
Data	`sahal42/ke-meter-readings`, pre-cropped LCD images with readings: 4,248 train, 1,214 validation, 606 test; every image has a reading
Length	50 epochs, 132 steps each, about 6,600 steps; 11–12 minutes per model
Validation	the 1,214 validation crops are read after every epoch (exact match and CTC loss); the best epoch is saved separately but not used
Model kept	final epoch, 50; the best-validation checkpoints score the same within one crop (vote 98.0% against 98.2%)
Test crops	scored once per epoch for the log only; never used to choose anything
Five runs	shift · seed 0, v1 · seed 0, shift · seed 1, shift · seed 2, v1 · seed 1
Result	606 test crops: single models 96.7–97.7%, five-model vote 98.2%

EACH TRAINING SAMPLE

1. A pre-cropped LCD image from the dataset, in grey.
2. Box jitter, applied as a warp of the crop: centre ±3% of the width and ±6% of the height, width × 0.95–1.12, height × 0.92–1.15, angle ±2°, edges extended; then resized to 64 × 256.
3. Lighting: brightness × 0.7–1.3, offset ±25, gamma 0.7–1.4; blur σ 0.5–1.2 on 30%, noise σ 3–10 on 30%.
4. `shift` only: moiré (35%, amplitude 10–45, period 2.5–7 px), fading (35%, contrast × 0.3–0.7), low resolution (30%, scale 0.35–0.7), motion blur (15%, 3–7 px), JPEG (30%, quality 15–50).
5. Grey copied to 3 channels, ImageNet normalisation.
6. Target: the reading's digits without leading zeros; digit d is class d + 1, class 0 is the CTC blank.

OPTIMISATION, THE SAME FOR EVERY RUN

Loss	CTC, blank 0
Optimiser	AdamW, lr 5e-4, weight decay 1e-4
Schedule	one-cycle: 10% warm-up from 2e-5 to 5e-4, then cosine decay to about 2e-9
Batch	32, shuffled, incomplete last batch dropped
Precision	fp16 mixed precision with gradient scaling; gradient clipping at norm 5
Start	ResNet18 layers ImageNet-pretrained; BiLSTM and linear layer from scratch